

Quiz 1

Aug 30th, 2024

Time: 12 minutes

Name: _____ Roll no.: _____ Group: _____

Instructions: Notes, books, computers, cell phones and other electronic devices are not allowed.
Max marks = 5.

Problem 1. 1. Find the reduced row echelon form of the augmented matrix of the system of equations:

$$\frac{2}{3}x_1 - \frac{8}{3}x_2 - \frac{2}{3}x_3 + \frac{2}{3}x_4 = 2$$

$$2x_1 - 8x_2 + x_3 - 4x_4 = 9$$

$$-x_1 + 4x_2 - 2x_3 + 5x_4 = -6$$

2. Is the above system consistent? (Give reasons for your answer)
3. Write the solution(s) of the above system in set notation.
4. How many solutions are there?